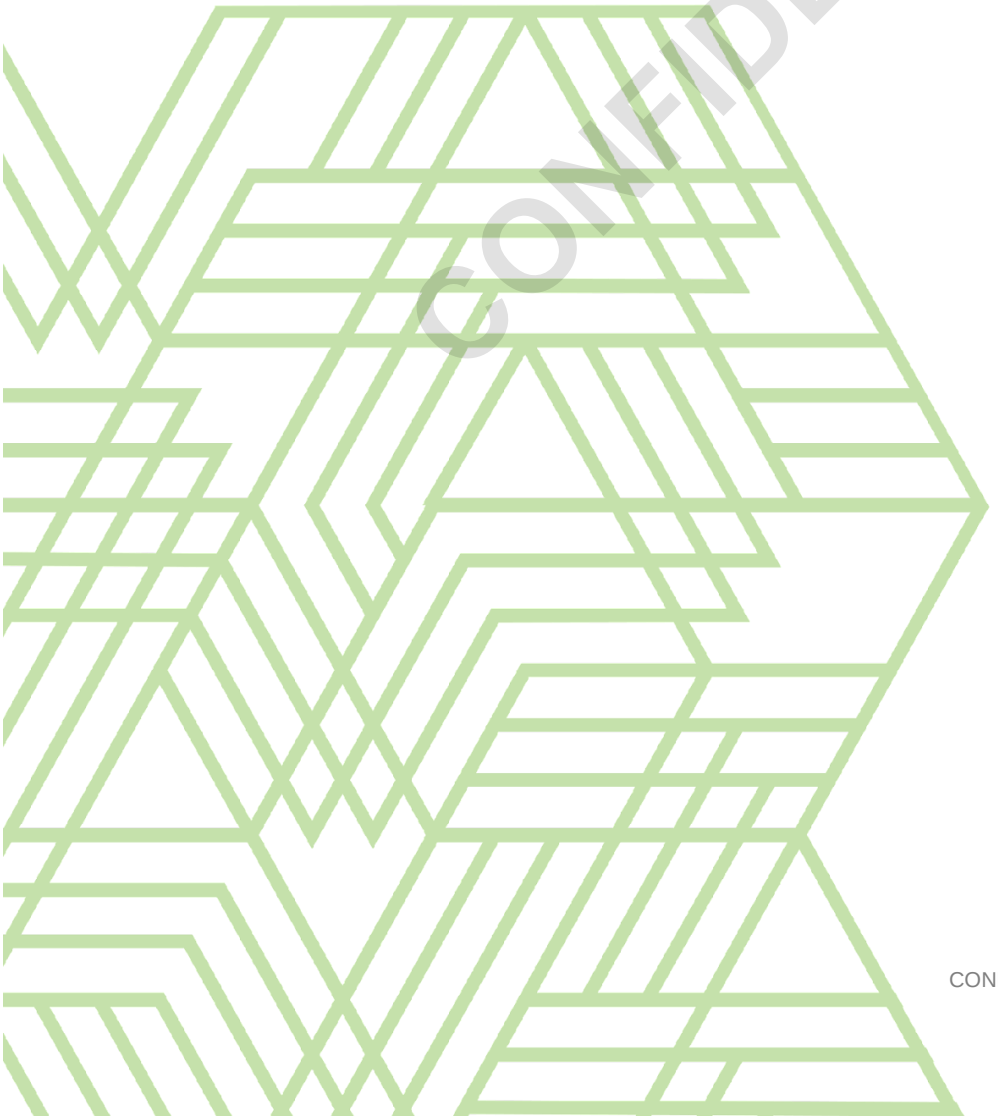
ATM33/e Series MCUboot

Application Note

SUMMARY: This document describes the MCUboot bootloader operation of the ATM33/ATM33e Wireless SoC Series.

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1. Overview

This Application note describes MCUboot bootloader operation and its usage. Document also covers MCUboot handling of Over The Air (OTA) image and security.

1.1 Introduction

MCUboot bootloader is an open source secure bootloader for 32 bit microcontrollers. ATM33/e uses this bootloader to support OTA operation and secure boot. Please refer to https://www.mcuboot.com/ for more details.

2. Building and Using MCUboot

MCUboot is required when OTA is enabled. One of the main functions of the MCUboot is to provide upgrade services to swap and validate an image to check for data corruption.

To enable MCUboot, run the following command at the application example directory of your design (e.g., platform/BLE_adv):

```
$ make clean run_all USE_MCUBOOT=1 ATMWSTK=PD100
```

- ATMWSTK=PD100 is set to configure the memory layout for Bluetooth stack. ATMWSTK specifies version of the Atmosic ATM Bluetooth Wireless Stack.
- USE_MCUBOOT=1 is used to enable MCUboot in the boot chain.
    - MCU bootloader is built using examples/ATM_mcuboot folder in SDK.

3. MCUboot RRAM Partitioning

When using MCUboot, RRAM partitioning information can be visualized using layout_info make target. When using SEGGER/Keil/IAR IDE build environment, similar layout can be seen using the IDE tool provided in the SDK. For more details, refer to the IDE Auxiliary Flash Programming Tool User Guide available on the Atmosic Support website.

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Figure 1 - MCUboot RRAM Partitioning

Partition sizes can be adjusted using various compile time options. Please refer to examples/README file for further details.

4. MCUboot Boot Flow

MCUboot device boot flow is shown in Figure 2.

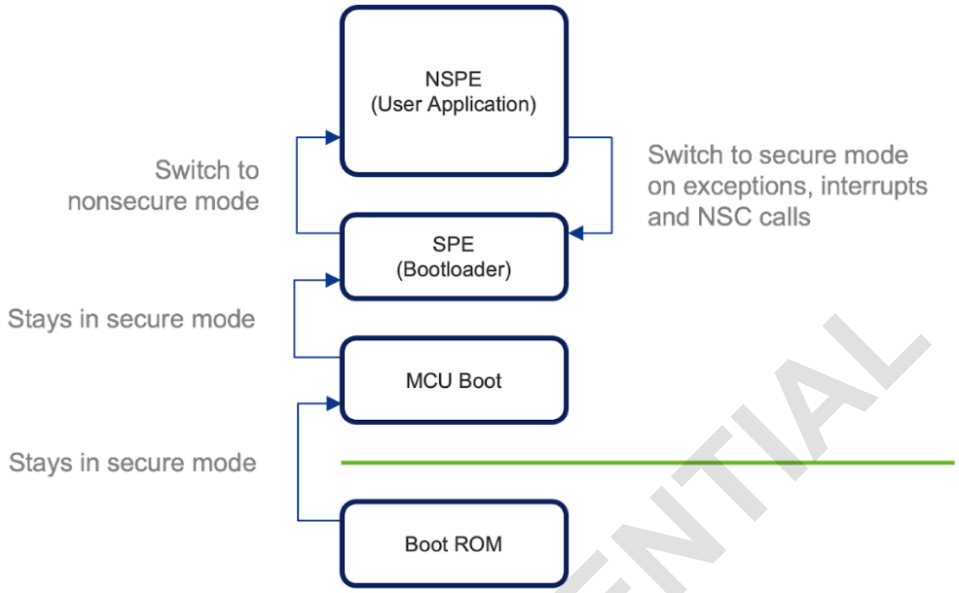


Figure 2 - MCUboot Device Boot Flow

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5. Secure Boot

Secure boot is a verification mechanism for ensuring that the new firmware is from a trusted origin. This is achieved by signing images with an ECDSA signature.

All examples can be built with secure boot enabled, run the following command:

```
$ make clean all USE_MCUBOOT=1 ATMWSTK=PD100 USE_SECURE_BOOT=1
```

USE_SECURE_BOOT=1 is used to enable secure boot. For more information on the design of the bootloader, please refer to https://docs.mcuboot.com/design.html

When signing images using USE_SECURE_BOOT, MCUboot uses a well known development private/public key as defined in contrib/mcuboot/root-ec-p256.pem file. Given this is a well known key in open source, it should not be used in production builds.

5.1 Key Generation

- Generating Public & Private key

  The key is used to sign the new image and it needs to be generated in a secure environment.

  To generate a new ecdsa_key.pem file, use the make target below from your specific application example/<APP> folder:

```
$ make gen_secure_boot_cert USE_MCUBOOT=1 ATMWSTK=PD100
```

  Private and public keys can be derived using commands below.

```
$  <SDK_ROOT>/contrib/mcuboot/scripts/imgtool.py getpub -k ecdsa_key.pem
$ <SDK_ROOT>/contrib/mcuboot/scripts/imgtool.py getpriv -k ecdsa_key.pem
```

- Public key

  The public key is used to verify the signed image. The public key will be embedded in the MCUboot image during compilation.

- Building and programming

  The new image signing process and the compiling MCUboot with generated public key is all included when running the following command:

```
$ make clean all ATMWSTK=PD100 USE_MCUBOOT=1 USE_SECURE_BOOT=1
SECURE_BOOT_KEY=ecdsa_key.pem
```

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- To program

```
$ make run all ATMWSTK=PD100 USE_MCUBOOT=1 USE_SECURE_BOOT=1
SECURE_BOOT_KEY=ecdsa_key.pem
```

5.2 Image Signing

Image signing is done during the build process, by generating signature pair (r,s) using a random number, selected elliptic curve domain parameters, private key and image hash (SHA256). See Figure 3.

Build process combines secure and nonsecure binaries (i.e. bootloader.elf and <app>.elf) into a single image before signing.

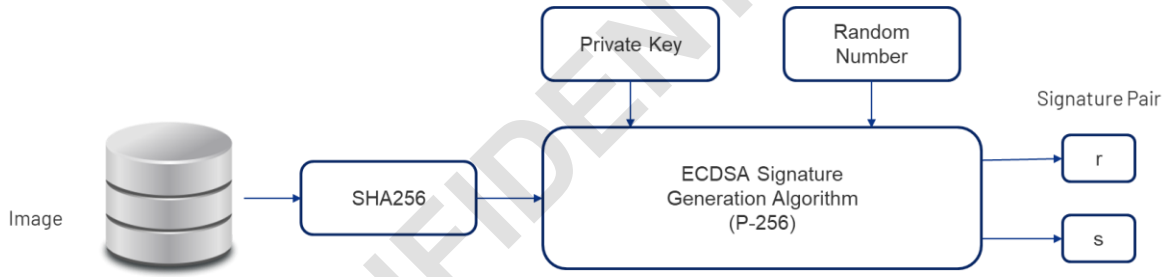


Figure 3 - Image Signing Build Process

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5.3 Image Verification

To verify the image, use the signature pair and public key stored in the image. See Figure 4.

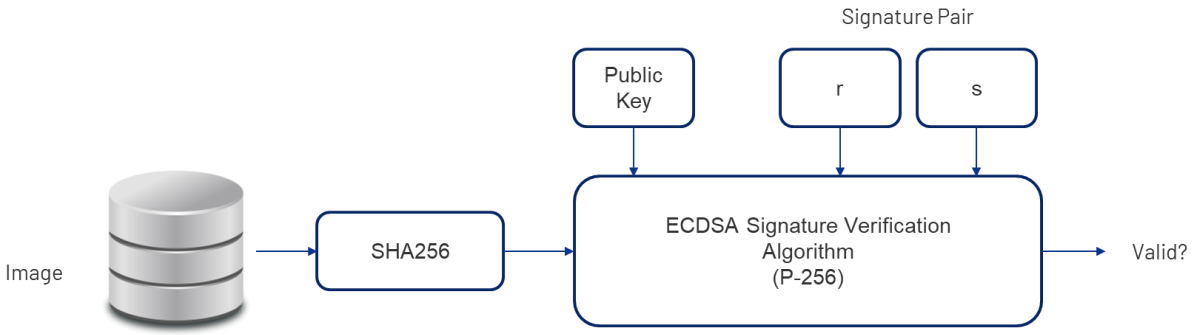


Figure 4 - Image Verification Process

5.4 Image format

Final image generated from the build process (signed_<APP>.ubin) has the following format.



Figure 5 - Image Format

Image signed_<APP>.bin follows the above format, but does not include padding and image trailer.

Image Header:

```
/** Image header.  All fields are in little endian byte order. */
struct image_header {
    uint32_t ih_magic;
    uint32_t ih_load_addr;
    uint16_t ih_hdr_size;           /* Size of image header (bytes). */
    uint16_t ih_protect_tlv_size;   /* Size of protected TLV area (bytes). */
    uint32_t ih_img_size;           /* Does not include header. */
    uint32_t ih_flags;              /* IMAGE_F_[...]. */
    struct image_version ih_ver;
    uint32_t _pad1;
};
```

***Note**: Refer to contrib/mcuboot/boot/bootutil/include/bootutil/image.h*

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TLV:

```
/** Image TLV header.  All fields in little endian. */
struct image_tlv_info {
    uint16_t it_magic;
    uint16_t it_tlv_tot;  /* size of TLV area (including tlv_info header) */
};

/** Image trailer TLV format. All fields in little endian. */
struct image_tlv {
    uint16_t it_type;   /* IMAGE_TLV_[...]. */
    uint16_t it_len;    /* Data length (not including TLV header). */
};
```

***Note**: Refer to contrib/mcuboot/boot/bootutil/include/bootutil/image.h*

Image Trailer:

```
*   0                   1                   2                   3
*   0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  ~                                                               ~
*  ~    Swap status (BOOT_MAX_IMG_SECTORS * min-write-size * 3)    ~
*  ~                                                               ~
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |                Encryption key 0 (16 octets) [*]               |
*  |                                                               |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |                Encryption key 1 (16 octets) [*]               |
*  |                                                               |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |                     Swap size (4 octets)                      |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |   Swap info   |          0xff padding (7 octets)              |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |   Copy done   |          0xff padding (7 octets)              |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |   Image OK    |          0xff padding (7 octets)              |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*  |                      MAGIC (16 octets)                        |
*  |                                                               |
*  +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
*
```

***Note**: Refer to contrib/mcuboot/boot/bootutil/src/bootutil_priv.h*

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MAGIC (16 octet) field refers to below:

```
const uint32_t boot_img_magic[] = {
    0xf395c277,
    0x7fefd260,
    0x0f505235,
    0x8079b62c,
};
```

Note: *Refer to https://docs.mcuboot.com/design.html and contrib/mcuboot/boot/bootutil/src/bootutil_public.c*

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References

Title	Document Number	Link
Atmosic SDK Quick Reference Guide	ATM-QRGSDK	Atmosic Support website, submit a support request for access credentials if needed.
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT	Atmosic Support website
MCUboot bootloader design	N/A	https://docs.mcuboot.com/design.html

Revision History

Date	Version	Description
November 11, 2022	0.50	Initial version created.

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